

# Flow-Factory OPD & Ensemble Eval Experiments

Flow-Factory

Generated 2026-05-22 09:31 UTC from W&B project Flow-Factory-OPD-Experiments

## 1 Experimental Setup

Offline `ensemble-eval` on SD3.5 Medium: FlowGRPO LoRA checkpoints are fused each denoising step, then evaluated on `geneval`, `pickscore`, and `ocr` held-out sets. Checkpoints: `jieliu/SD3.5M-FlowGRPO-Text` (OCR), `jieliu/SD3.5M-FlowGRPO-PickScore`, `jieliu/SD3.5M-FlowGRPO-GenEval`. Multi-teacher rows use AVG unless marked PCGRAD. Table 1: **bold** = single-teacher in-domain; gray = multi-teacher in-domain; red/blue = change vs. max single-teacher baseline in that row. OPD rows (*peak*) use the maximum eval logged so far.

## 2 Main Results

Table 1: Cross-benchmark `eval/` means. Top: offline ensemble-eval (last eval). Bottom: OPD training runs still **running** (peak eval). **Bold**: single-teacher in-domain.

| Config.                                   | Agg.        | GenEval benchmark |             | PickScore benchmark | OCR benchmark |             |
|-------------------------------------------|-------------|-------------------|-------------|---------------------|---------------|-------------|
|                                           |             | GenEval           | PickScore   | PickScore           | OCR           | PickScore   |
| <i>Ensemble eval (offline, last eval)</i> |             |                   |             |                     |               |             |
| SD3.5 (base)                              | —           | 0.657             | 0.875       | 0.835               | 0.574         | 0.866       |
| geneval                                   | —           | <b>0.937</b>      | 0.888       | 0.829               | 0.656         | 0.862       |
| ocr                                       | —           | 0.685             | 0.876       | 0.837               | <b>0.931</b>  | 0.869       |
| pickscore                                 | —           | 0.641             | 0.918       | <b>0.895</b>        | 0.690         | 0.915       |
| geneval+ocr                               | AVG         | 0.850-0.087       | 0.888-0.000 | 0.837+0.000         | 0.875-0.055   | 0.874+0.005 |
| geneval+pickscore                         | AVG         | 0.830-0.107       | 0.908-0.010 | 0.873-0.021         | 0.675-0.015   | 0.899-0.017 |
| ocr+pickscore                             | AVG         | 0.680-0.005       | 0.901-0.017 | 0.873-0.022         | 0.865-0.066   | 0.899-0.016 |
| geneval+ocr+pickscore                     | PCGRAD      | 0.794-0.143       | 0.901-0.017 | 0.864-0.031         | 0.844-0.087   | 0.894-0.022 |
| <i>OPD training (running, peak eval)</i>  |             |                   |             |                     |               |             |
| pw-avg                                    | AVG peak    | 0.787-0.150       | 0.898-0.020 | 0.861-0.034         | 0.830-0.101   | 0.893-0.023 |
| pw-sum                                    | SUM peak    | 0.744-0.193       | 0.895-0.023 | 0.857-0.037         | 0.793-0.138   | 0.891-0.025 |
| pw-sum-rbs                                | SUM peak    | 0.810-0.127       | 0.889-0.029 | 0.874-0.020         | 0.884-0.046   | 0.883-0.032 |
| v-pcgrad                                  | PCGRAD peak | 0.808-0.129       | 0.889-0.029 | 0.872-0.022         | 0.873-0.058   | 0.884-0.032 |
| v-pcg-nr                                  | PCGRAD peak | 0.747-0.190       | 0.896-0.022 | 0.857-0.038         | 0.787-0.144   | 0.890-0.025 |
| diff-opd-rbs                              | D-OPD peak  | 0.909-0.028       | 0.892-0.026 | 0.886-0.008         | 0.911-0.019   | 0.884-0.032 |

## 3 Full Eval Metrics

Table 2: All scalar `eval/` metrics (std and GenEval per-tag). Ensemble: last eval; OPD: peak.

| Config.                                   | Agg.        | GenEval benchmark |            |        |          |          |            |           |       |            |        |          |          | PickScore benchmark |         | OCR benchmark |       |       |
|-------------------------------------------|-------------|-------------------|------------|--------|----------|----------|------------|-----------|-------|------------|--------|----------|----------|---------------------|---------|---------------|-------|-------|
|                                           |             | GenEval           |            |        |          |          |            | PickScore |       |            |        |          |          | PickScore           | OCR     | PickScore     |       |       |
|                                           |             | Std               | Color attr | Colors | Counting | Position | Single obj | Two obj   | Std   | Color attr | Colors | Counting | Position | Single obj          | Two obj | Std           | Std   | Std   |
| <i>Ensemble eval (offline, last eval)</i> |             |                   |            |        |          |          |            |           |       |            |        |          |          |                     |         |               |       |       |
| SD3.5 (base)                              | —           | 0.395             | 0.521      | 0.711  | 0.761    | 0.450    | 0.844      | 0.636     | 0.042 | 0.900      | 0.879  | 0.875    | 0.866    | 0.855               | 0.873   | 0.057         | 0.393 | 0.039 |
| geneval                                   | —           | 0.205             | 0.854      | 0.867  | 0.994    | 0.950    | 1.000      | 0.982     | 0.038 | 0.920      | 0.890  | 0.890    | 0.869    | 0.864               | 0.887   | 0.056         | 0.346 | 0.038 |
| ocr                                       | —           | 0.375             | 0.531      | 0.689  | 0.802    | 0.490    | 0.867      | 0.664     | 0.042 | 0.902      | 0.882  | 0.879    | 0.874    | 0.856               | 0.874   | 0.057         | 0.173 | 0.041 |
| pickscore                                 | —           | 0.407             | 0.479      | 0.622  | 0.913    | 0.440    | 0.800      | 0.655     | 0.043 | 0.949      | 0.919  | 0.917    | 0.926    | 0.881               | 0.921   | 0.058         | 0.350 | 0.039 |
| geneval+ocr                               | Avg         | 0.307             | 0.719      | 0.800  | 0.898    | 0.730    | 0.911      | 0.945     | 0.040 | 0.921      | 0.886  | 0.889    | 0.880    | 0.867               | 0.884   | 0.056         | 0.231 | 0.040 |
| geneval+pickscore                         | Avg         | 0.319             | 0.740      | 0.800  | 0.904    | 0.770    | 0.933      | 0.873     | 0.041 | 0.947      | 0.907  | 0.907    | 0.906    | 0.875               | 0.909   | 0.057         | 0.349 | 0.038 |
| ocr+pickscore                             | Avg         | 0.387             | 0.490      | 0.711  | 0.867    | 0.550    | 0.778      | 0.682     | 0.042 | 0.935      | 0.897  | 0.903    | 0.905    | 0.869               | 0.900   | 0.057         | 0.253 | 0.040 |
| geneval+ocr+pickscore                     | PCGRAD      | 0.338             | 0.625      | 0.844  | 0.959    | 0.650    | 0.867      | 0.818     | 0.041 | 0.940      | 0.899  | 0.898    | 0.898    | 0.870               | 0.900   | 0.058         | 0.259 | 0.039 |
| <i>OPD training (running, peak eval)</i>  |             |                   |            |        |          |          |            |           |       |            |        |          |          |                     |         |               |       |       |
| pw-avg                                    | Avg peak    | 0.393             | 0.677      | 0.800  | 0.957    | 0.760    | 0.889      | 0.873     | 0.042 | 0.938      | 0.897  | 0.900    | 0.900    | 0.869               | 0.898   | 0.058         | 0.393 | 0.040 |
| pw-sum                                    | SUM peak    | 0.395             | 0.604      | 0.800  | 0.933    | 0.710    | 0.867      | 0.773     | 0.042 | 0.932      | 0.895  | 0.897    | 0.896    | 0.869               | 0.894   | 0.058         | 0.388 | 0.040 |
| pw-sum-rbs                                | SUM peak    | 0.395             | 0.740      | 0.889  | 0.922    | 0.720    | 0.889      | 0.827     | 0.042 | 0.923      | 0.898  | 0.893    | 0.883    | 0.864               | 0.878   | 0.059         | 0.388 | 0.042 |
| v-pcgrad                                  | PCGRAD peak | 0.395             | 0.729      | 0.822  | 0.913    | 0.700    | 0.911      | 0.836     | 0.042 | 0.924      | 0.899  | 0.891    | 0.882    | 0.863               | 0.881   | 0.059         | 0.388 | 0.042 |
| v-pcg-ur                                  | PCGRAD peak | 0.395             | 0.573      | 0.800  | 0.926    | 0.680    | 0.844      | 0.782     | 0.042 | 0.933      | 0.894  | 0.896    | 0.894    | 0.869               | 0.897   | 0.058         | 0.388 | 0.040 |
| diff-opd-rbs                              | D-OPD peak  | 0.389             | 0.833      | 0.889  | 0.970    | 0.890    | 0.956      | 0.955     | 0.042 | 0.925      | 0.900  | 0.895    | 0.885    | 0.870               | 0.890   | 0.059         | 0.388 | 0.041 |